C64 PSU GUARD – protected 5 V / 12 V feed for a 1541-II / 1581 drive from a PC power supply.
v0.2 2026-10-26: J1 PSU in = JST XH B4B-XH-A, keyed (1 +5V, 2/3 GND, 4 +12V; Berg <-> XH adapter cable)
-> J2 drive out (DIN pin 1 = 5V, 2 = GND, 4 = 12V).
Channel 5 V: limit 1.2 A, OVP 6.0 V, UVLO 4.2 V. Channel 12 V: limit 1.5 A, OVP 14.0 V, UVLO 10.4 V.
TPS25921A = auto retry: red LED on while the channel is off (thermal shutdown), off again when the fault is gone.
J3 UFI link = JST XH B4B-XH-A, 1:1 XH cable to UFI J20 GUARD: 1 GND, 2 OFF (high = both channels off), 3 ERR5, 4 ERR12.

5 V channel: reverse blocking P-MOSFET (body diode conducts first, then Vgs = -Vin, zener clamps the gate), TVS + input capacitors, TPS25921A eFuse.
UVLO divider 100k/49.9k, OVP divider 100k/30k, R_ILIM 115k (I_LIMIT ~ 0.0105 A/kOhm, fast trip 0.0142 A/kOhm + 0.36 A), C_SS 22 nF.

+5V_IN
+12V_IN
J1
PSU_IN
Q1
SI2401A
G5
V5_P
D1
BZT52C10
D2
SMF5.0CA
100k R1
C1
10uF/35V
C2
100nF
30k R5
49.9k R3
C3
22nF
SS5
OV5
EN5
100k R4
100k R2
U1
TPS25921ADR
IN
ENUV
OVP
SS
GND
OUT
FLT
ILIM
+5V_OUT
FLT5
IL5
115k R6
C4
1uF
Q2
2N7002
OFF
1k R7
D3
green
LG5
V5_RG5
4.7k R8
D4
red
LR5

12 V channel: reverse blocking P-MOSFET (body diode conducts first, then Vgs = -Vin, zener clamps the gate), TVS + input capacitors, TPS25921A eFuse.
UVLO divider 100k/15.4k, OVP divider 100k/11k, R_ILIM 143k (I_LIMIT ~ 0.0105 A/kOhm, fast trip 0.0142 A/kOhm + 0.36 A), C_SS 22 nF.

+12V_IN
Q3
SI2401A
G12
V12_P
D5
BZT52C10
D6
SMAJ13A
100k R9
C5
10uF/35V
C6
100nF
11k R13
15.4k R11
C7
22nF
SS12
OV12
EN12
100k R12
100k R10
U2
TPS25921ADR
+12V_OUT
FLT12
IL12
143k R14
C8
1uF
Q4
2N7002
3.3k R15
D7
green
LG12
V12_RG12
4.7k R16
D8
red
LR12

Reverse polarity LED: conducts only when +5V_IN or +12V_IN is negative against GND
(1N4148W blocks the rail voltage in normal operation).

2.2k R17
REVA
D9
red
REV
D10
1N4148W
D11
1N4148W

J4 external fault LED: 1 = anode (+), 2 = cathode (-); on while the 5 V or the 12 V channel is off.

UFI link: OFF high = both channels off. ERR5/ERR12 = open drain, released (high) when the
channel reports a fault or the guard is unpowered, pulled low while the channel is on.
UFI Headless v1.0 J20 GUARD (1:1 XH cable): 1 GND, 2 OFF <- PG7 (220R), 3 ERR5 -> PG8, 4 ERR12 -> PC9
(10k pull-ups to 3.3 V, 220R series resistors and PESD3V3L1BA on the UFI side).

J2
DRIVE
J3
UFI_LINK
Q5
2N7002
Q6
2N7002
100k R18

Pull-ups (v0.2): 100k FLT5 -> V5_P, 100k FLT12 -> V12_P (TPS25921A FLT: open drain, 20 V max).

D12
1N4148W
XK

UFI Project
Sheet: /
File: C64_PSU_Guard.kicad_sch
Title: C64 PSU Guard
Size: A3 | Date: 2026-10-05 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 1/1